

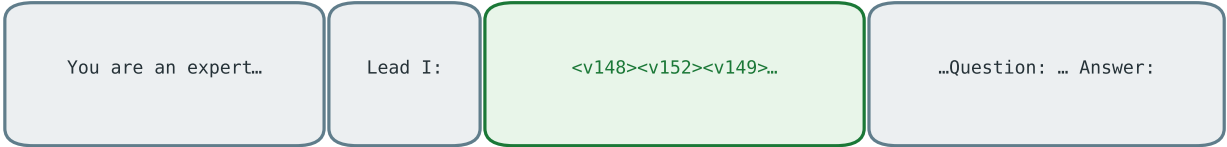
OpenTSLM-Mamba: a pure-token, fully state-space TSLM

no signal encoder · no projector · no task head — the only TS-specific parameters are 256 rows of the embedding table

* frozen

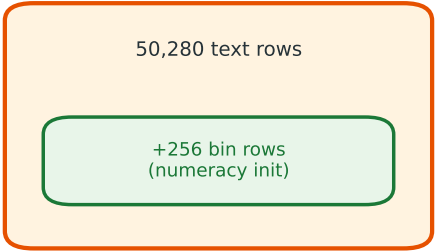
* trainable

Interleaved token stream (one sequence, one vocabulary)



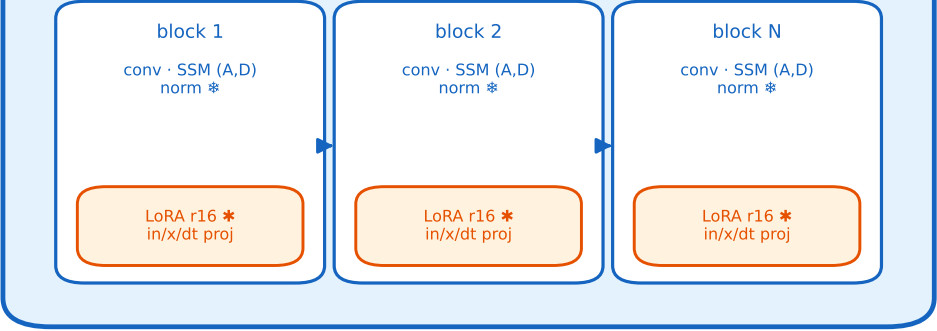
values → bin tokens (z-score, 256 uniform bins)

Shared embedding table *



tied with LM head

Mamba backbone * (xN blocks, linear-time selective scan, constant-size state)



1.37 B frozen · 118 M trainable (1.4b) | 59 M trainable (370m)

LM head *
(tied)

decode constrained to bin
tokens (logit mask)

Text out

"(b)" · CoT
rationales

Signal out

<v149><v151>...
→ dequantize
median-of-20 @ T=.5

Training recipe (both phases train the SAME parameters: LoRA + embeddings)

① Signal-native pretraining (≤1 GPU-h)

next-bin forecasting · corpus = 90% real (freq-balanced) / 10% synthetic GP
+3–4 pts at short budgets · removes seed-dependent seasonality failure

② Instruction tuning (OpenTSLM curriculum)

TSQA → captioning → CoT stages · LoRA r16, lr 1e-4, AdamW
right-pad train / left-pad generate (fused-scan padding rule)

Stage-1 TSQA (full test, protocol v4): OpenTSLM-Mamba-1.4b 0.9983 · paper-faithful OpenTSLM-SP 0.9946 — and only the token interface also generates signal.